

Charm4Py File Structure

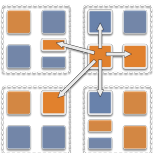
- At least 1 .py file
- Program entry point: main method
- Chares = Python classes that extend the Chare class

Hello World Example

- Running
 - `python3 -m charmrun.start +p4 array_hello.py`
 - The +p4 tells the system to use four cores

Starting the program

- `charm.start(main)`
 - Tells the runtime to start running the `main()` method on PE 0
 - Program setup in this method

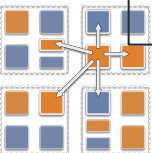


Hello World Example: main()

```
from charm4py import charm
```

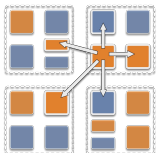
```
def main(args):  
    print("Hello world")  
    exit()
```

```
charm.start(main)
```



Chare creation

- Chares are Python classes that extend the Chare class
 - Chare is defined internally in charm4py module
- Same definition can be used for 3 types of chare structures
 - Singleton chares
 - Groups
 - Arrays



Hello World Example: Hello chare

```
from charm4py import charm, Chare

class Hello(Chare):
    def __init__(self):
        print("Hello from PE: " + str(charm.myPe()))
        exit()

def main(args):
    c = Chare(Hello, onPe=0)

charm.start(main)
```